11711BCC008
Matheus José da Costa

1)

```
[12/08/2020 04:55] seed@ubuntu:~/hash_criptografia_assimetrica$ cat arquivo
Matheus Jose da Costa 11711BCC008
```

b)

```
[12/08/2020 04:56] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl dgst -md
5 arquivo
MD5(arquivo)= f128ab1e10eaebba38f9530d3760ddd6
[12/08/2020 04:56] seed@ubuntu:~/hash_criptografia_assimetrica$ ls
arquivo
[12/08/2020 04:56] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl dgst sha
256 arquivo
sha256: No such file or directory
MD5(arquivo)= f128ab1e10eaebba38f9530d3760ddd6
[12/08/2020 04:57] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl dgst -sh
a256 arquivo
SHA256(arquivo)= f3ea28f3957e478b579e0b86c577434f407abddce87b7c853d124b5b10e7de8
3
[12/08/2020 04:57] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl dgst -sh
a512 arquivo
SHA512(arquivo)= 7845c9b848846c32b1c594221d9b853807a71d4f74cae69cab63e4f10fb807e
ae3e8e463de2398e2a1b95d6c64c7976ee8851172cb7aa36af9f654d4fdc762d8
[12/08/2020 04:57] seed@ubuntu:~/hash_criptografia_assimetrica$
```

c) Algoritmos usados: MD5, SHA256, SHA512

2)

a)

```
[12/08/2020 05:00] seed@ubuntu:~/hash_criptografia_assimetrica$ touch exercicio2
```

b)

```
[12/08/2020 05:01] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl dgst -md
5 exercicio2
MD5(exercicio2)= e7dc19f00ff774c1fafa85ffa8e4b5ce
```

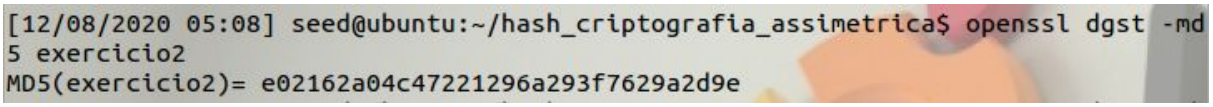
c)

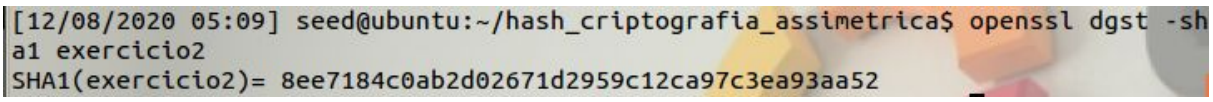
```
[12/08/2020 05:01] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl dgst -sh
a1 exercicio2
SHA1(exercicio2)= e189528d415b4b91650c2a6d2bc2da8cda598283
```

d)



e)



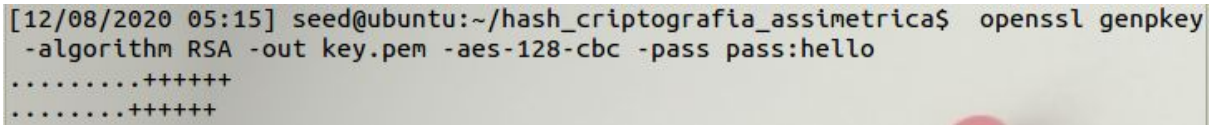


f)

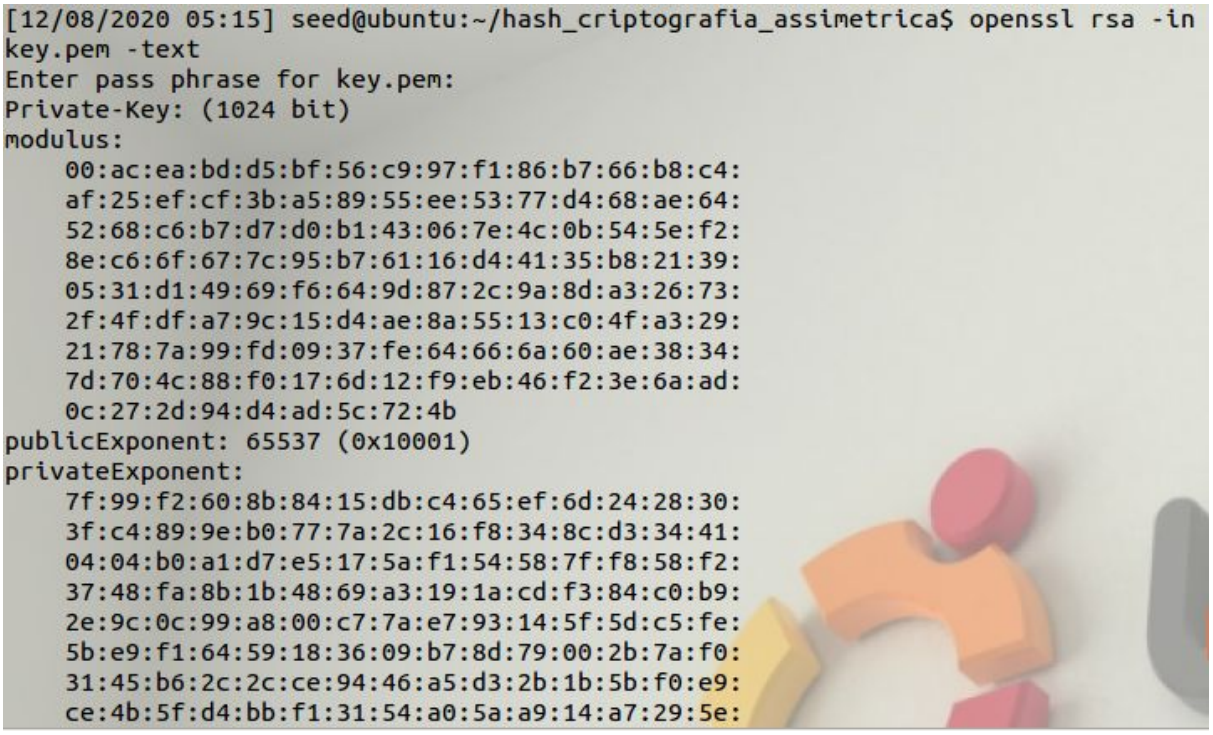
Os hashs são bastante parecidos.

3-

a)



b) **openssl rsa -in key.pem -text**



```
    34:b5:9a:85:b8:90:57:85:81:3f:1e:82:52:5d:7c:
    8d:7a:04:09
writing RSA key
-----BEGIN RSA PRIVATE KEY-----
MIICXAIBAAKBgQCs6r3Vv1bJl/GGt2a4xK8l7887pYlV7lN31GiuZFJoxrfX0LFD
Bn5MC1Re8o7Gb2d8lbdhFtRBNbghOQUx0Ulp9mSdhyyajaMmcy9P36ecFdSuilUT
wE+jKSF4epn9CTf+ZGZqYK44NH1wTIjwF20S+etG8j5qrQwnLZTUrVxySwIDAQAB
AoGAf5nyYIuEFdvEZe9tJCgwP8SJnrB3eiwW+DSM0zRBBASwodflF1rxVFh/+Fjy
N0j6ixtIaaMZGs3zhMC5LpwMmagAx3rnkxRfXcX+W+nxZFkYNgm3jXkAK3rwMUW2
LCzOlEal0ysbW/Dpzktf1LvxMVSgWqkUpyle5Ta9vaJgeBECQQDgvSLMlaV2fTM5
0iffzHFpL/evPJzG88040R4kjJmKHUIY1ja0zYZ1ca1uhbncd9g5UZWZhVObTH2G
UulpeTs9AkEAxPg/wFfkShb1WEbW916BW1Kz5won2F9AsL3iD0r7wEMpL0vCTYEE
ASQNM6Cl3pwEp5XagCQyG6LwA+s+IKzcJwJBAIm3XPvBOAPD9QAZqBWcUmMC4eSg
X/IKtFd8zdvXzL617ysJGbzxtNODNaPkjmBrkDSPqfzJltIzrTgC1sWrZIECQF0d
ltXRNyUncVkINDvBO5n2oQHKNUf+z9A+FA5yvk0H8wJe6D5HjaMUbgFIIuH4E+M9
3ovRkLar8peHsFk2HuUCQCtX82h4Ew+JClBGwoK75EH4uE/ss+FtM6WlGaj+lv8c
iZku4O5xa4QN7YW7HjS1moW4kFeFgT8eglJdfI16BAk=
-----END RSA PRIVATE KEY-----
```

openssl rsa -in key.pem -pubout -text

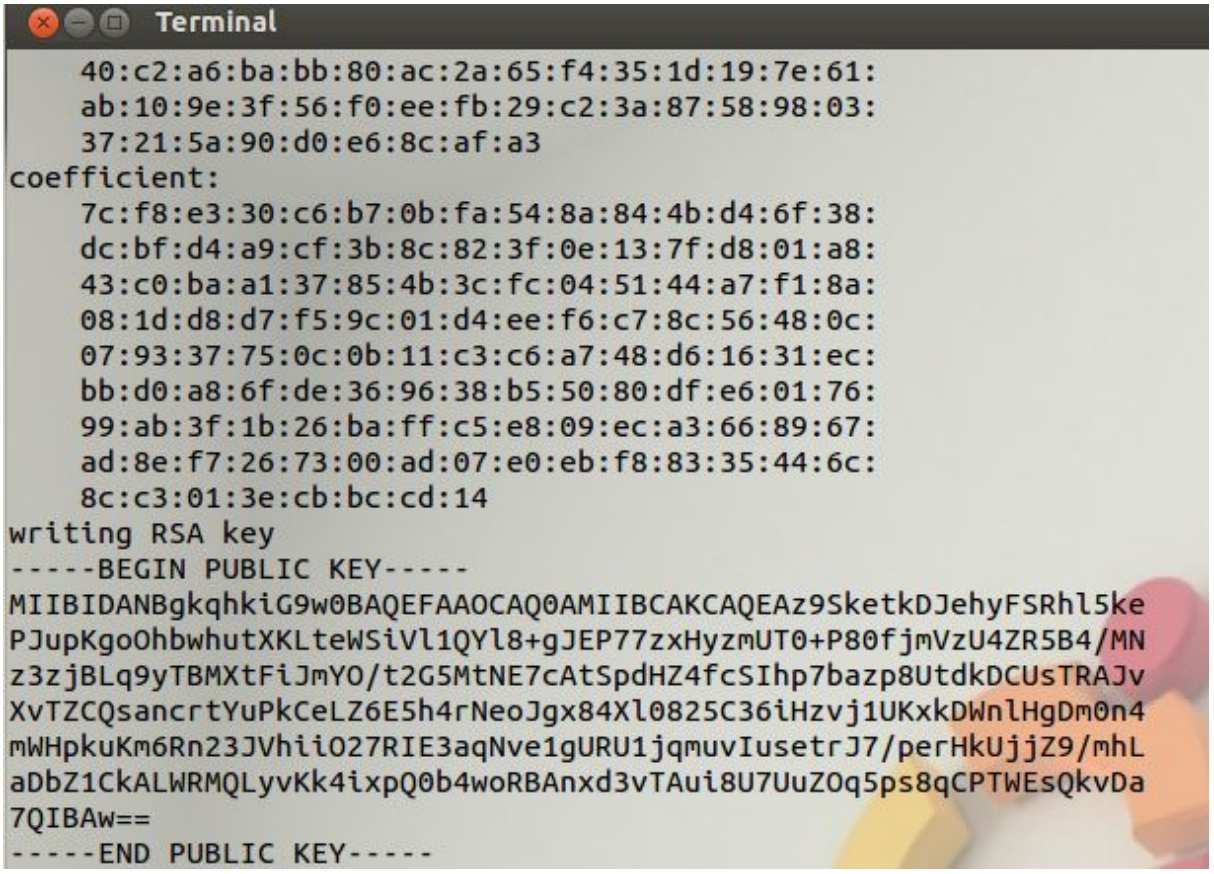
```
[12/08/2020 05:20] seed@ubuntu:~/hash_criptografia_assimetrica$ openssl rsa -in
key.pem -pubout -text
Enter pass phrase for key.pem:
Private-Key: (1024 bit)
modulus:
    00:ac:ea:bd:d5:bf:56:c9:97:f1:86:b7:66:b8:c4:
    af:25:ef:cf:3b:a5:89:55:ee:53:77:d4:68:ae:64:
    52:68:c6:b7:d7:d0:b1:43:06:7e:4c:0b:54:5e:f2:
    8e:c6:6f:67:7c:95:b7:61:16:d4:41:35:b8:21:39:
    05:31:d1:49:69:f6:64:9d:87:2c:9a:8d:a3:26:73:
    2f:4f:df:a7:9c:15:d4:ae:8a:55:13:c0:4f:a3:29:
    21:78:7a:99:fd:09:37:fe:64:66:6a:60:ae:38:34:
    7d:70:4c:88:f0:17:6d:12:f9:eb:46:f2:3e:6a:ad:
    0c:27:2d:94:d4:ad:5c:72:4b
publicExponent: 65537 (0x10001)
privateExponent:
    7f:99:f2:60:8b:84:15:db:c4:65:ef:6d:24:28:30:
    3f:c4:89:9e:b0:77:7a:2c:16:f8:34:8c:d3:34:41:
    04:04:b0:a1:d7:e5:17:5a:f1:54:58:7f:f8:58:f2:
    37:48:fa:8b:1b:48:69:a3:19:1a:cd:f3:84:c0:b9:
    2e:9c:0c:99:a8:00:c7:7a:e7:93:14:5f:5d:c5:fe:
    5b:e9:f1:64:59:18:36:09:b7:8d:79:00:2b:7a:f0:
    31:45:b6:2c:2c:ce:94:46:a5:d3:2b:1b:5b:f0:e9:
    ce:4b:5f:d4:bb:f1:31:54:a0:5a:a9:14:a7:29:5e:
```

```
    52:63:02:e1:e4:a0:5f:f2:0a:b4:57:7c:cd:db:d7:
    cc:be:b5:ef:2b:09:19:bc:f1:b4:d3:83:35:a3:e4:
    8e:60:6b:90:34:8f:a9:fc:c9:96:d2:33:ad:38:02:
    d6:c5:ab:64:81
exponent2:
    5d:1d:96:d5:d1:37:25:27:71:59:08:34:3b:c1:3b:
    99:f6:a1:01:ca:35:47:fe:cf:d0:3e:14:0e:72:be:
    4d:07:f3:02:5e:e8:3e:47:8d:a3:14:6e:01:48:22:
    e1:f8:13:e3:3d:de:8b:d1:90:b6:ab:f2:97:87:b0:
    59:36:1e:e5
coefficient:
    2b:57:f3:68:78:13:0f:89:0a:50:46:c2:82:bb:e4:
    41:f8:b8:4f:ec:b3:e1:6d:33:a5:a5:19:a8:fe:96:
    ff:1c:89:99:2e:e0:ee:71:6b:84:0d:ed:85:bb:1e:
    34:b5:9a:85:b8:90:57:85:81:3f:1e:82:52:5d:7c:
    8d:7a:04:09
writing RSA key
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCs6r3Vv1bJl/GGt2a4xK8l7887
pYlV7lN31GiuZFJoxrfX0LFDBn5MC1Re8o7Gb2d8lbdhFtRBNbghOQUx0Ulp9mSd
hyyajaMmcy9P36ecFdSuilUTwE+jKSF4epn9CTf+ZGZqYK44NH1wTIjwF20S+etG
8j5qrQwnLZTUrVxySwIDAQAB
-----END PUBLIC KEY-----
```

c)
key.pem

```
Terminal
    b7:50:20:45:d2:04:9d:11:53:b5:af:d5:51:fb:67:
    2f:d1:17:d4:a1:ed:a5:f0:72:6e:0a:65:db:cb:68:
    79:f3:b7:35:8b:8c:ef:3d:a2:9c:1c:4a:a0:e1:83:
    2b:1d:43:04:85
exponent2:
    2d:ea:62:5f:db:a3:94:fb:01:ca:96:45:b3:ce:3d:
    19:42:45:17:8d:80:33:33:a7:05:65:dc:4d:57:85:
    0c:53:38:c1:d5:5a:34:f1:f6:9e:43:22:7d:2b:f6:
    4d:98:74:19:32:49:c0:9c:36:11:66:74:ce:26:44:
    53:da:b5:1d
coefficient:
    69:cd:7b:16:29:34:8b:16:68:28:2a:5c:a0:83:c2:
    7b:34:44:6b:1f:17:a2:7e:6d:a2:d3:d5:d6:d5:8f:
    92:6a:97:6f:8b:ff:47:5c:a0:1d:26:3f:56:b4:ff:
    34:39:45:0e:d9:7c:17:2c:2e:ce:ef:39:5f:ec:09:
    0e:61:04:00
writing RSA key
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDOJKQYePhiXCVvAKHydJ9MwTF3
eS/uibG9M5pqvwHrCjOgYt+ejzJTULdH4vPVVN2wKknw2kHVgHCO8Qm359RIt8MK
ZTXTd+RDanJeC8U1rkJZQFJ0odL8LgGY6zVZwF54nmRG8GWUzLVbKldbx9Z4zLm2
GnzUWsTAJMLkMRKccQIDAQAB
-----END PUBLIC KEY-----
[12/08/2020 05:33] seed@ubuntu:~/hash_criptografia_assimetrica$
```

key1.pem


Terminal
40:c2:a6:ba:bb:80:ac:2a:65:f4:35:1d:19:7e:61:
ab:10:9e:3f:56:f0:ee:fb:29:c2:3a:87:58:98:03:
37:21:5a:90:d0:e6:8c:af:a3
coefficient:
7c:f8:e3:30:c6:b7:0b:fa:54:8a:84:4b:d4:6f:38:
dc:bf:d4:a9:cf:3b:8c:82:3f:0e:13:7f:d8:01:a8:
43:c0:ba:a1:37:85:4b:3c:fc:04:51:44:a7:f1:8a:
08:1d:d8:d7:f5:9c:01:d4:ee:f6:c7:8c:56:48:0c:
07:93:37:75:0c:0b:11:c3:c6:a7:48:d6:16:31:ec:
bb:d0:a8:6f:de:36:96:38:b5:50:80:df:e6:01:76:
99:ab:3f:1b:26:ba:ff:c5:e8:09:ec:a3:66:89:67:
ad:8e:f7:26:73:00:ad:07:e0:eb:f8:83:35:44:6c:
8c:c3:01:3e:cb:bc:cd:14
writing RSA key
-----BEGIN PUBLIC KEY-----
MIIBIDANBgkqhkiG9w0BAQEFAAOCAQ0AMIIBCAKCAQEAz9SketkDJehyFSRhl5ke
PJupKgoOhbwhutXKLteWSiVl1QYl8+gJEP77zxHyzmUT0+P80fjmVzU4ZR5B4/MN
z3zjBLq9yTBMXtFiJmYO/t2G5MtNE7cAtSpdHZ4fcSIhp7bazp8UtdkDCUsTRAJv
XvTZCQsancrtYuPkCeLZ6E5h4rNeoJgx84Xl0825C36iHzvj1UKxkDWnlHgDm0n4
mWHpkuKm6Rn23JVhiiO27RIE3aqNve1gURU1jqmuvIusetrJ7/perHkUjjZ9/mhL
aDbZ1CkALWRMQLyvKk4ixpQ0b4woRBAnxd3vTAui8U7UuZOq5ps8qCPTWEsQkvDa
7QIBAw==
-----END PUBLIC KEY-----